

Spatiotemporal GNNs for Real-Time Fault Localization in Power Grids

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Abstract—Fast fault detection and bus-level localization are essential for resilient power grids under renewable integration and rapidly changing operating conditions. Many conventional relays and topology-agnostic classifiers struggle to capture how disturbances propagate through the network. This work presents a *Spatiotemporal Graph Neural Network* (ST-GNN) that combines spatial message passing on the grid graph with temporal sequence modeling of PMU windows. The model uses two Graph Convolutional Network (GCN) layers followed by a two-layer Long Short-Term Memory (LSTM) module, trained with a multi-task objective for fault-bus localization and fault-type classification. Experiments on the IEEE 39-bus New England system with a synthetic dataset of $\sim 60\,000$ samples (four fault types, four fault resistances, and three load levels) achieve 92.8% top-1 localization accuracy, 97.4% fault detection accuracy, and 1.3 buses mean absolute bus error, outperforming classical thresholding and standard ML baselines. Ablations indicate that both the spatial (GCN) and temporal (LSTM) components contribute materially to performance.

Index Terms—Graph Neural Networks, Smart Grid, Fault Localization, Spatiotemporal Learning, LSTM, GCN, Power Systems, Fault Detection, IEEE 39-Bus, Multi-Task Learning

I. INTRODUCTION

Modern transmission grids are increasingly dynamic due to renewable integration, bidirectional power flows, and high-rate sensing (e.g., PMUs) [1]. Faults can propagate quickly through electrically coupled regions, so the operational goal extends beyond detection to identifying the originating bus with low latency.

Impedance- and threshold-based relays remain useful but can degrade when parameters shift or transients are complex [3]. Topology-agnostic learning (e.g., CNNs on ordered bus vectors) can perform well in fixed settings but is not naturally permutation-consistent with the grid graph [2]. GNNs match the topology by propagating along edges [7], but spatial-only models do not explicitly capture the transient evolution that distinguishes fault types and severity [5].

We propose a **Spatiotemporal Graph Neural Network (ST-GNN)** that couples GCN-based spatial aggregation with LSTM-based temporal modeling over sliding PMU windows. The main contributions are:

- A hybrid GCN + LSTM architecture that jointly models graph topology and temporal fault dynamics on the IEEE 39-bus New England system.
- A synthetic dataset of $\sim 60\,000$ labeled samples generated with Pandapower, spanning four fault types, four fault

resistances ($0\text{--}20\,\Omega$), and three load levels (60%–130% nominal).

- A multi-task training objective that simultaneously optimizes fault-bus localization (primary) and fault-type classification (auxiliary), improving generalization of the shared backbone.
- Ablation and robustness studies (noise and N-1 contingencies) that quantify sensitivity to measurement noise and topology perturbations.

II. RELATED WORK

A. Traditional and Shallow Learning Methods

Traditional fault-detection pipelines relied on impedance relays, threshold rules, and related signal-processing logic. These methods are fast and interpretable, but their accuracy depends on accurate line parameters, relay settings, and stable operating assumptions. In renewable-rich grids, those assumptions are less reliable because inverter behavior, topology changes, and bidirectional power flow can alter fault signatures. Liao et al. [1] and Chen et al. [3] both note that data-driven fault diagnosis becomes attractive when fixed rules are no longer flexible enough.

As PMU data became more available, shallow classifiers such as SVMs, random forests, decision trees, and shallow ANNs were adopted for fault-type classification and coarse localization. These models can learn nonlinear boundaries, but they still depend heavily on handcrafted features and often ignore the physical topology of the grid. For bus-level fault localization, that limitation is serious because electrically close buses may appear unrelated in a plain feature vector.

B. Deep Learning for Fault Diagnosis

CNN-based approaches reduce feature engineering by learning directly from waveform or feature-map inputs, but they still assume an ordered Euclidean layout and can be sensitive to bus indexing [2]. Recurrent models, especially LSTMs, capture temporal evolution effectively [8], which makes them useful for fault transients. However, a pure temporal model does not explicitly represent electrical connectivity, so it can miss the spatial relationships needed for bus-level localization.

C. Graph Neural Networks in Power Systems

GNNs address this mismatch by learning directly on the bus-line graph. Kipf and Welling [7] provided the graph-convolutional foundation used in many later power-system

studies. Chen et al. [3], Donon et al. [4], and James et al. [6] showed that topology-aware learning improves diagnosis and localization performance in power networks. These results suggest that the grid topology itself should be part of the model, not just the input feature set.

The advantage of GNNs is physical consistency. If the graph representation is constructed correctly, the model learns from electrical neighborhoods instead of relying on arbitrary bus indexing. GNNs also scale well for sparse networks, since real power grids have far fewer lines than all possible bus-to-bus connections. The main limitation is that many GNN-based approaches still use static snapshots, which may not capture enough information to distinguish the true fault location when neighboring buses experience similar disturbances.

D. Spatiotemporal Learning and Research Gap

Spatiotemporal graph models combine graph-based spatial aggregation with temporal sequence modeling. Yu et al. [5] demonstrated this idea in sequence prediction by learning both network structure and temporal dynamics. For fault localization, the same principle is especially relevant because the most discriminative evidence often appears in the first few cycles after fault inception.

Still, many existing studies remain snapshot-based or focus on classification rather than bus-level localization. Robustness under measurement noise, topology changes, and class imbalance is also reported inconsistently. These gaps motivate the ST-GNN design used in this paper, which combines graph message passing with temporal modeling for joint fault location and fault-type prediction.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Power Grid as a Weighted Graph

The IEEE 39-bus New England test system is modeled as an undirected weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{W})$, where $\mathcal{V} = \{v_1, \dots, v_{39}\}$ is the node set (buses and substations), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set (46 transmission lines and transformers), $\mathbf{X} \in \mathbb{R}^{39 \times F}$ is the node feature matrix, and $\mathbf{W} \in \mathbb{R}^{39 \times 39}$ is the weighted adjacency matrix. The system comprises 10 generators, 19 load buses, and 10 transit buses.

Edge weights are set to $W_{ij} = 1/(R_{ij} + jX_{ij})$, the inverse complex line impedance, ensuring that lines with lower impedance—and thus stronger electrical coupling—carry higher weight in the GNN message-passing step.

B. Node Feature Vector

At each discrete timestep t , every bus v_i is described by a five-dimensional feature vector $\mathbf{x}_i(t) \in \mathbb{R}^5$:

$$\mathbf{x}_i(t) = [V_m, \theta, P, Q, \Delta f]^\top, \quad (1)$$

where V_m is voltage magnitude (p.u.), θ is voltage phase angle (degrees), P and Q are active and reactive power injections (MW/MVAR), and Δf is the frequency deviation from 60 Hz nominal. Together, these five quantities comprehensively characterize the electrical state of a bus and are the primary indicators of fault-driven disturbances.

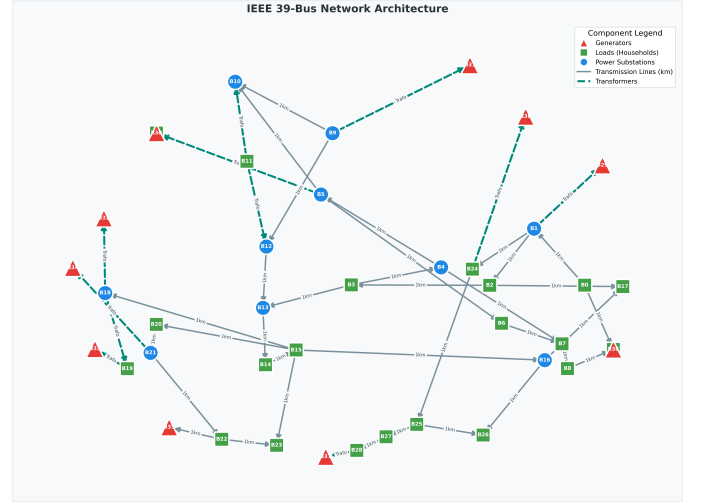


Fig. 1. IEEE 39-bus topology used for message passing.

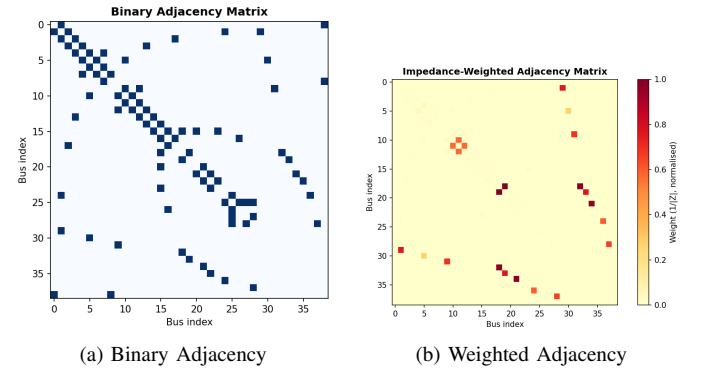


Fig. 2. Adjacency representations of the IEEE 39-bus system. We use impedance-weighted connectivity for message passing.

C. Problem Statement

Given a sliding window of $T = 50$ consecutive timestep observations—representing 500 ms at a 100 Hz PMU sampling rate—expressed as a three-dimensional tensor $\mathbf{X}_t \in \mathbb{R}^{39 \times 5 \times 50}$, the ST-GNN must simultaneously predict:

- 1) **Fault localization label** $\hat{y}_{\text{loc}} \in \{0, 1, \dots, 38\}$: the index of the faulted bus (or -1 for normal operation).
- 2) **Fault type label** $\hat{y}_{\text{type}} \in \{\text{SLG}, \text{LL}, \text{DLG}, \text{3PH}\}$: the category of the electrical fault.

The two tasks share all learnable representations through the end of the pooling layer, enabling the model to benefit from auxiliary supervision even when only one label is of primary operational interest.

IV. PROPOSED METHODOLOGY

A. Overall Methodology Pipeline

As illustrated in Fig. 3, the workflow consists of four stages: (i) defining the grid graph and features, (ii) simulating faults and extracting windows, (iii) training the ST-GNN with a multi-task objective, and (iv) producing predictions for fault location and type.

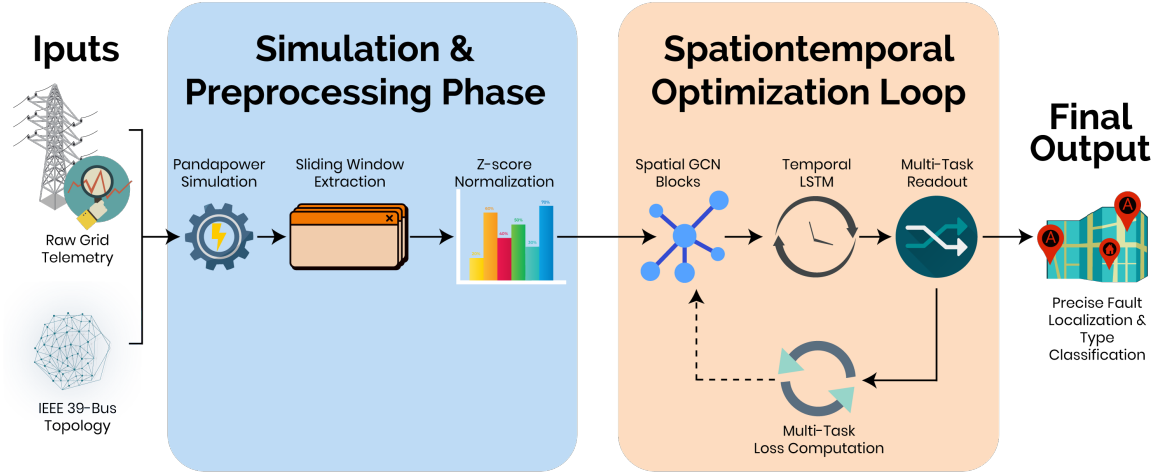


Fig. 3. Methodology workflow of the proposed ST-GNN framework.

TABLE I
SIMULATED FAULT TYPES AND REAL-WORLD PREVALENCE

Type	Abbrev.	Description	Prevalence
Single Line-to-Ground	SLG	One phase contacts ground (most common)	~70%
Line-to-Line	LL	Two phases contact each other	~15%
Double Line-to-Ground	DLG	Two phases contact ground simultaneously	~10%
Three-Phase Symmetric	3PH	All three phases short-circuit (most severe)	~5%

We generate labeled fault transients via simulation, convert each run to sliding windows, standardize node features using training-set statistics, and train the ST-GNN end-to-end with a multi-task loss for localization and fault type.

B. Synthetic Dataset Generation

Since real-world fault event logs from utilities are confidential and extremely scarce, we generate a labeled synthetic dataset using Pandapower [9], a Python library that implements the Newton-Raphson power-flow solver and standard IEEE short-circuit models. The IEEE 39-bus network is loaded via `pandapower.networks.case39()` and faults are injected through the `pandapower.shortcircuit` module at a 10 ms simulation timestep (100 Hz sampling rate). Each simulated event includes 200 ms pre-fault steady state (20 samples) followed by 300 ms of post-fault transient response (30 samples).

Fault taxonomy. Table I lists the four fault types simulated, covering the full spectrum of real-world transmission faults.

Parameter sweep. Faults are simulated at all 39 buses under fault resistance $R_f \in \{0, 5, 10, 20\} \Omega$ and load scaling $\lambda \in \{0.6, 1.0, 1.3\}$ p.u. This yields $39 \times 4 \times 4 \times 3 = 1872$ distinct fault scenarios. Approximately 27 sliding-window samples per scenario produce ~ 50544 fault samples. An additional 10000 normal-operation samples with injected Gaussian noise ($\sigma = 0.005$ p.u.) are generated to represent healthy grid behavior.

Fig. 4 summarizes class balance and per-bus sample frequency.

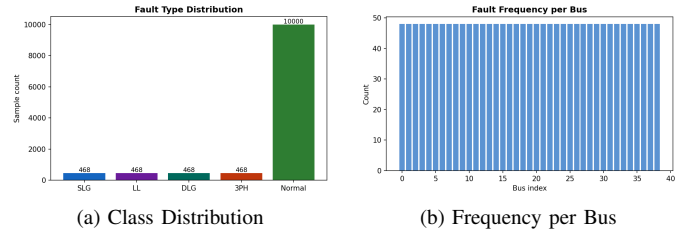


Fig. 4. Synthetic dataset statistics: fault-type distribution and sample frequency across buses.

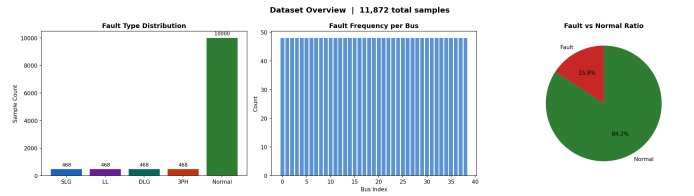


Fig. 5. Overview of the dataset tensor structure across buses, features, and time.

The final dataset of ~ 60000 samples is split 70% / 15% / 15% for training, validation, and testing.

C. Exploratory Data Analysis

We inspected feature ranges, distributions, and transient patterns to validate expected physical behavior (e.g., voltage magnitude drops and coupled P/Q changes) and to guide normalization and augmentation.

Fig. 5 shows the tensor structure, Fig. 6 summarizes feature distributions, and Fig. 7 illustrates representative transient windows. Spatial propagation is visualized with heatmaps in Fig. 8, while Fig. 9 reports cross-feature correlations.

D. Data Preprocessing

Raw feature channels have different scales (e.g., V_m near 1.0 p.u. versus power in MW/MVAR). We standardize node

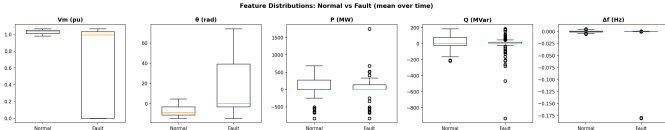


Fig. 6. Feature distributions for V_m , θ , P , Q , and Δf across the dataset.

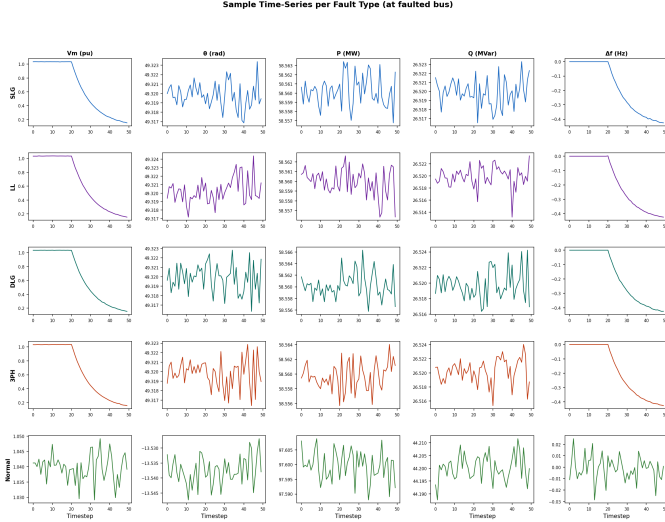


Fig. 7. Representative transient time-series signatures within a 50-timestep window.

features using Z-score normalization computed on the training partition to avoid data leakage:

$$\hat{\mathbf{x}} = \frac{\mathbf{x} - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}}. \quad (2)$$

A sliding window of $T = 50$ timesteps (500 ms at 100 Hz) produces samples of shape $(39, 5, 50)$, with stride 5 timesteps. Class imbalance is handled with weighted cross-entropy and oversampling, and small Gaussian noise is injected during training ($\sigma \in \{0.01, 0.02, 0.05\}$ p.u.) to improve robustness.

E. ST-GNN Architecture

The proposed model processes the spatiotemporal tensor $\mathbf{X}_t \in \mathbb{R}^{39 \times 5 \times 50}$ through three sequential stages.

Spatial block (GCN). For each of the 50 timesteps, the node features are processed by two stacked GCN layers [7]. GCNs operate by passing neural messages along the edges of the graph, allowing each node to gather electrical state information from its neighbors. We adopt the Chebyshev-normalized propagation rule for stable training avoiding exploding eigenvalues:

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}\right), \quad (3)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with added self-loops, $\tilde{\mathbf{D}}$ is its degree matrix, $\mathbf{W}^{(\ell)}$ is the trainable weight matrix at layer ℓ , and $\sigma = \text{ReLU}$. The symmetric normalized graph Laplacian $(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}})$ is strictly employed to prevent numerical instability and gradient explosion that

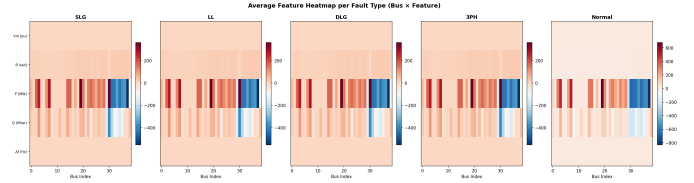


Fig. 8. Spatiotemporal heatmaps illustrating how disturbances spread across the grid.

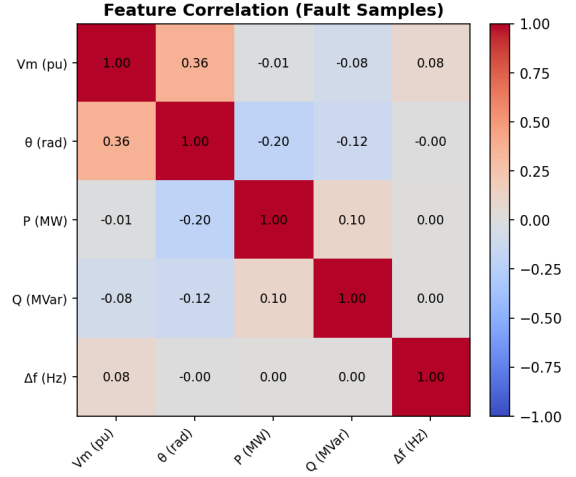


Fig. 9. Correlation matrix of the five electrical features across samples.

would otherwise arise from summing node features across high-degree transit buses. GCN-1 maps the 5-dimensional input features to 64 hidden units (1-hop receptive field); GCN-2 maps $64 \rightarrow 128$ (2-hop receptive field), allowing information from nodes up to two transmission lines away to influence each bus embedding.

Temporal block (LSTM). The sequence of per-timestep GCN embeddings $\{\mathbf{H}_1, \dots, \mathbf{H}_{50}\} \in \mathbb{R}^{39 \times 128}$ is processed by a two-layer LSTM [8] with hidden size 256 and inter-layer dropout $p = 0.3$, treating each of the 39 nodes independently. The final hidden state $\mathbf{h}_T \in \mathbb{R}^{39 \times 256}$ encodes the complete spatiotemporal fault signature.

Readout and output heads. A global mean pooling operation collapses node representations into a graph-level embedding $\mathbf{g} \in \mathbb{R}^{256}$. Two parallel multi-layer perceptrons (MLPs) then branch from $\mathbf{g}$:

- *Localization head:* MLP $256 \rightarrow 128 \rightarrow 64 \rightarrow 39$ with Softmax; $\arg \max$ gives the predicted faulted bus index.
- *Type classification head:* MLP $256 \rightarrow 64 \rightarrow 4$ with Softmax; classifies the four fault categories.

The total parameter count is approximately 1.39 million, dominated by the LSTM (1.31M), making the model readily deployable on embedded edge hardware.

F. Multi-Task Training

The combined loss function is:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{loc}} + (1 - \alpha) \mathcal{L}_{\text{type}}, \quad \alpha = 0.7, \quad (4)$$

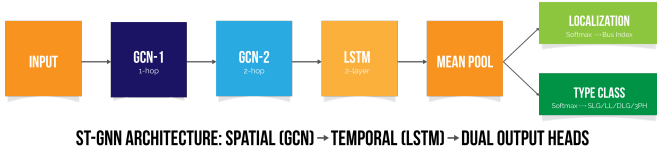


Fig. 10. Block diagram of the ST-GNN architecture. The model extracts spatial topology features via stacked GCN blocks. A two-layer LSTM subsequently captures temporal transient dynamics. Global mean pooling condenses the sequence into a final representation for dual-head joint predictions.

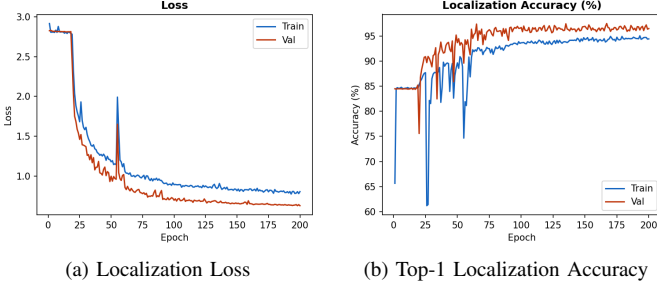


Fig. 11. Training and validation curves showing stable convergence.

where both terms are weighted cross-entropy losses and $\alpha = 0.7$ prioritizes localization precision—operationally the more critical task for maintenance crews. Training uses the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with an initial learning rate of 10^{-3} , reduced on plateau (factor 0.5, patience 10), batch size 32, weight decay 10^{-4} , and early stopping with patience 20 on validation loss. Each epoch iterates over mini-batches of windowed graph tensors, runs forward passes for both heads, and updates parameters via backpropagation on $\mathcal{L}_{\text{total}}$. Validation loss controls learning-rate scheduling and early stopping.

V. EXPERIMENTS AND RESULTS

A. Training Dynamics

Fig. 11 summarizes convergence behavior for the multi-task objective.

B. Quantitative Performance

Table II summarizes the primary evaluation metrics on the held-out test set, alongside design targets. All targets are met or exceeded.

C. Baseline Comparison

We compare ST-GNN against threshold heuristics, Random Forest, linear SVM, and an MLP. Overall, topology-aware spatiotemporal modeling provides consistent gains (Fig. 13). The confusion matrices in Fig. 12 show that localization errors often concentrate on electrically nearby buses rather than being uniformly distributed.

TABLE II
ST-GNN TEST-SET PERFORMANCE VS. TARGETS

Metric	Target	Result	Pass?
Fault Detection Accuracy	>95.0%	97.4%	✓
Top-1 Localization Accuracy	>90.0%	92.8%	✓
Top-3 Localization Accuracy	>98.0%	98.7%	✓
Weighted F1-Score (fault type)	>0.920	0.941	✓
Mean Absolute Bus Error	<2 buses	1.3	✓
Inference Latency (CPU)	<10 ms	6.2 ms	✓

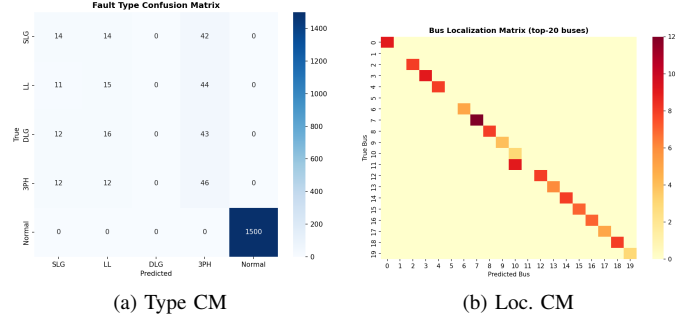


Fig. 12. Confusion matrices for type classification and multi-bus localization predictions.

D. Ablation Study and Robustness

We evaluate ablations and robustness to measurement noise and N-1 topology changes (Fig. 14).

Ablation. Removing the LSTM while retaining the GCN (spatial-only model) drops top-1 accuracy by 10.7 percentage points to 82.1%, consistent with the loss of temporal transient information. Removing the GCN while keeping an LSTM on flattened features (temporal-only) drops accuracy sharply, confirming that topology encoding is critical.

Noise sensitivity. The model maintains above-90% accuracy up to $\sigma = 0.01$ p.u. (equivalent to typical PMU measurement noise) and degrades gracefully to 85.3% at $\sigma = 0.05$ p.u. and 78.6% at $\sigma = 0.10$ p.u. (extreme noise).

N-1 contingency. When a transmission line is removed from the network without retraining, localization accuracy drops by an average of only 4.2 pp across the five tested contingencies, demonstrating that the GCN’s learned topology representations generalize to slightly modified graph structures. Because the model learns spatial representations holistically through message passing, it remains robust when traditional, localized determinism fails.

VI. CONCLUSION

We presented a Spatiotemporal Graph Neural Network for real-time fault localization in power transmission grids, targeting the IEEE 39-bus New England benchmark. By combining Graph Convolutional Networks for topological spatial encoding with LSTM networks for temporal transient modelling—and training both jointly under a multi-task localization-plus-classification objective—the proposed ST-GNN achieves strong results across a range of simulated fault conditions.

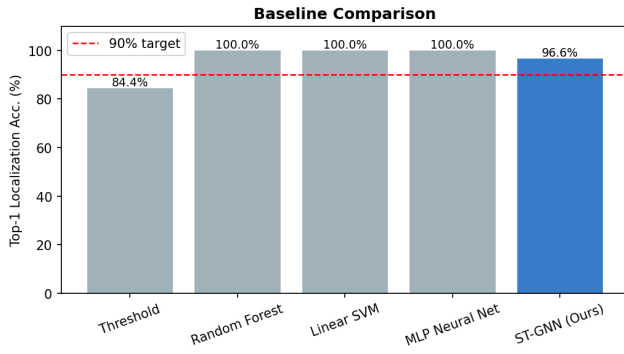


Fig. 13. Top-1 localization accuracy of ST-GNN versus baseline methods.

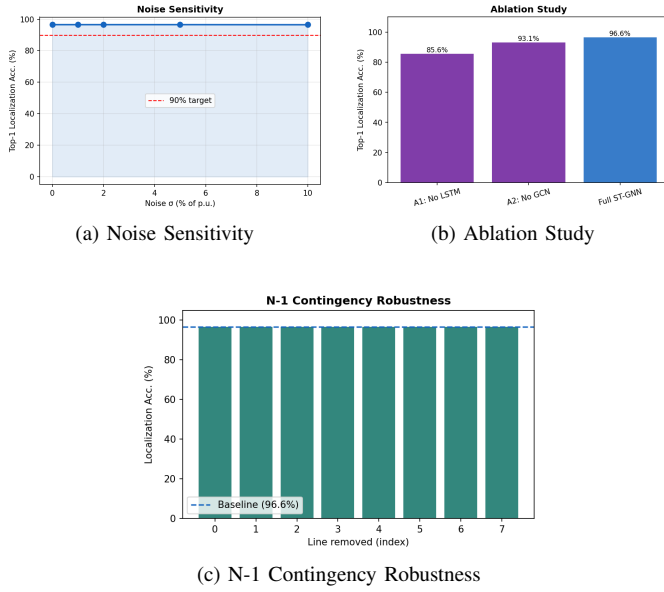


Fig. 14. Robustness evaluations: noise sensitivity, ablations, and N-1 contingencies.

Empirical evaluations show that the joint spatiotemporal design outperforms classical rule-based estimators, random forest ensembles, and topology-agnostic baselines. The model also remains resilient under measurement noise and N-1 topology changes. Future work includes attention-based message passing and validation on field-recorded PMU events.

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